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Distributions of Candidate Vaccine Targets, Virulence Factors, and Resistance Features of Invasive Group B Streptococcus Using Whole-Genome Sequencing: A Multicenter, Population-based Surveillance Study

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Highlights:

- Largest GBS genome sequence sample of Chinese infants.
- Baseline data for GBS surveillance and vaccine impact evaluation.

ABSTRACT

Background. Group B *Streptococcus* (GBS) is a leading cause of morbidity and mortality in young infants worldwide. This study aimed to investigate candidate GBS vaccine targets, virulence factors, and antimicrobial resistance determinants.

Methods. We used whole-genome sequencing to characterize invasive GBS isolates from infants <3 months of age obtained from a multicenter population-based study conducted from 2015 to 2021 in China.

Results. Overall, seven serotypes were detected from 278 GBS isolates, four (Ia, Ib, III, V) of which accounted for 97.8%. We detected 30 sequence types (including 10 novel types) that were grouped into six clonal complexes (CCs: CC1, CC10, CC17, CC19, CC23 and CC651); three novel ST groups in CC17 were detected, and the rate of CC17, considered a hyperinvasive neonatal clone complex, was attached to 40.6% (113/278). A total of 98.9% (275/278) of isolates harbored at least one alpha-like protein gene. All GBS isolates contained at least one of three pilus backbone determinants and the pilus types *PI-2b* and *PI-1+PI-2a* accounted for 79.8% of the isolates. The 112 serotype III/CC17 GBS isolates were all positive for *hvgA*. Most of the isolates (75.2%) were positive for serine-rich repeat glycoprotein determinants (*srr1* or *srr2*). Almost all isolates possessed *cfb* (99.6%), *c1IE* (100%), *lmb* (95.3%) or *pavA* (100%) *gene.* Seventy-seven percent of isolates harboured more than three antimicrobial resistance genes with 28.4% (79/278) *gyr*A quinolone resistance determinants mutation, 83.8% (233/278) carrying *tet* cluster genes and 77.3% (215/278) carrying *erm* genes which mediated fluoroquinolone, tetracycline and clindamycin resistance, respectively."

Conclusions. The findings from this large whole-genome sequence of GBS isolates establish important baseline data required for further surveillance and evaluating the impact of future vaccine candidates.

KEYWORDS: Group B Streptococcus; whole-genome sequencing; surface protein; virulence factors; resistance genes.

1. Introduction

Group B *Streptococcus* (GBS) disease remains a leading contributor to adverse maternal and newborn outcomes[1, 2]. It is the most common cause of sepsis and meningitis in the first three months of life and accounts for a considerable disease burden worldwide, with the highest burden in sub-Saharan Africa and Asia[1]. Furthermore, a large proportion of GBS meningitis patients and sepsis survivors have long-term neurodevelopmental impairment [1]. A meta-analysis in 2017 among infants <3 months with invasive GBS disease estimated an overall global incidence of 0.49 per 1000 live births, with a case fatality risk of 8.4% [2]. In China, the estimated incidence of invasive GBS disease in infants was 0.31~0.55 per 1000 live births [3, 4].

Screening for rectovaginal GBS colonization coupled with intrapartum antibiotic prophylaxis (IAP) given to mothers at least four hours before delivery has reduced the incidence of early-onset disease (EOD; < 7 days of age) in many high-income countries[5]. Alternatively, IAP can be given to women with risk factors, as recommended by the Perinatal Medicine Branch of the Chinese Medical Association in 2021[6]. Nevertheless, the use of IAP is not effective against GBS late-onset disease (LOD; 7–89 days of age), and laboratory testing for GBS screening and the requirement to deliver antimicrobial agents are challenging in low-resource settings where the burden of GBS infection is highest[5]. Moreover, there are concerns about antibiotic resistance associated with the use of IAPs [7]. A high prevalence of inducible antimicrobial resistance to clindamycin and erythromycin has been detected among GBS isolates globally [3, 5, 8, 9].

Vaccine development efforts are being pursued to offer an alternative option to IAP for protection against invasive GBS disease [10]. A maternal GBS vaccine for pregnant women could result in the transplacental transfer of protective antibodies to the fetus and has the potential to reduce maternal disease, adverse pregnancy outcomes, and infant invasive disease[11-13]. The GBS vaccine could protect against both the EOD and LOD. The GBS vaccine is regarded as a priority vaccine for development by the World Health Organization (WHO) [14]. GBS vaccines currently in phase II clinical trials include serotype-specific capsular polysaccharide-protein conjugate and alpha-like protein-based vaccines, which are being evaluated in nonpregnant and pregnant women [15-17]. The GBS vaccine is likely to be cost-effective, with substantial health benefits and notable reductions in morbidity, mortality, and disability in children [18].

A number of virulence factors expressed by GBS, including capsular polysaccharide (CPS) and surface proteins, are involved in adherence, invasion, and immune evasion and are potential vaccine candidates [10]. Therefore, it is important to characterize the GBS structure better and understand its genetic lineages by using whole-genome sequencing (WGS) to define vaccine targets better. A study from the United States (US) analyzed GBS features by WGS and suggested that current GBS vaccine candidates have good coverage [19]. In the United Kingdom and Ireland, researchers have found that over half of genomic clusters were previously undetected, and the importance of routine genomic confirmation has been emphasized [20]. However, there is limited WGS data from China [21]. In this study, we aimed to define the clonal distribution of invasive GBS isolates, identify surface proteins that are potential vaccine epitopes, and uncover genetic determinants of antimicrobial resistance.

2. Methods

2.1 Study design and population

We conducted a multicenter, population-based surveillance study for invasive GBS disease among infants in 18 urban tertiary hospitals located in 16 provinces of China from January 2015 to December 2017, as described previously[4]. In addition, invasive GBS isolates were also collected from a selected hospital in Hubei Province between 2018 and 2021. A case of invasive GBS disease was defined as GBS isolated from a normally sterile site (blood, cerebrospinal fluid, or joint fluid) in infants younger than 3 months of age. All participating hospitals were provided a common, standardized clinical protocol to identify GBS patients. The criteria for hospital selection and recruitment processes have been previously described [4, 22]. Further details on invasive GBS disease identification, antimicrobial nonsusceptibility testing, microbiological culture, DNA extraction, and sequencing can be found in the Supplementary Material.

2.2 Molecular subtypes

Antibiotic resistance genes, serotypes, surface protein genes, and hypervirulent genes were determined using the GBS bioinformatics pipeline of the US Centers for Disease Control and Prevention (CDC) [23]. The ARG-ANNOT and ResFinder databases were used to detect additional resistance determinants [24, 25]. The surface protein genes encoding the hypervirulent GBS adhesin (*hvgA*), serine-rich repeat (*srr*) proteins, pilus islands (*PI-1, PI-1b, PI-2a1, PI-2a2, PI-2a3, PI-2a4, PI-2b, PI-2b2*), and alpha protein family (*alpha C, alp2/3, alp1* and *rib*) were extracted from the genomic data. For each sequence query, ≥95% identity was set as the threshold for a positive result. Moreover, the coding sequences (CDSs) of 11 surface proteins (*alp1, alp2/3, alpC, C5a, gbs2106, PI-1, PI-2a, PI-2b, rib, sip,* and *srr2)* were screened and analyzed in all 278 isolates to assess divergence and potential for vaccine candidates. The sequences of eleven genes were phylogenetically analyzed both individually and in tandem.

Multilocus sequence typing was performed using the allele profiles of 7 housekeeping genes (*adhP, pheS, atr, glnA, sdhA, glcK,* and *tkt*) to assign associated STs. STs were grouped into CCs. The eBURST algorithm on PubMLST was used to assess the founding sequence types and their respective CCs. The genomes of isolates with undiscovered sequence types were uploaded to the database to obtain new sequence types. BioNumerics 5.0 (Applied Maths, http://www.applied-maths.com) was used to create minimum spanning trees based on MLST.

2.3 Whole-Genome Single Nucleotide Polymorphism

We used MUMmer (version 3.23) to align and assemble the isolate sequences against the reference genome of BG-NI-011 using the default parameters and then used the delta–filter -r –q parameter with the one-to-one alignment block option to filter the alignment results. We used show-snp to identify SNPs. All SNPs of all isolates were merged as a matrix file. To improve the overall quality of the data, SNP positions with a distance of less than 5 bp, as well as positions that carry an unspecified nucleotide ("N"), were removed. Finally, we obtained an SNP sequence alignment with a length of 26101 bp for 278 isolates and 30529 bp for 333 isolates (including 278 from this study, 50 worldwide representative isolates that caused invasion infection, and 5 isolates representative of the 5 main cluster clones downloaded from PubMLST and NCBI). Phylogenetic reconstruction was performed with Fasttree (version 2.1.10) using the generalized time-reversible model, and the phylogenetic tree was visualized by ITOL (https://itol.embl.de).

2.4 Ethical approval

The study was approved by the Medical Ethics Committee of Guangzhou Women and Children's Medical Center, China. This Ethics Committee served as the central institutional review board for all facilities and approved this study (approval no. 2016050405). The permission of each participating hospital was obtained from their hospital or central institutional review board approval. We obtained written informed consent from the parents or guardians of infants with invasive GBS infection as described previously[4]. The GBS isolates in this study were obtained from our previously registered study in the US National Library of Medicine clinical trials database (http://clinicaltrials.gov) under registration no. NCT02812576.

2.5 Data availability

The complete genomic sequences of all isolates in our study were deposited at the China National Microbiology Data Center (NMDC, https://nmdc.cn/en) under BioProject Number 10018290.

3. Results

3.1 Serotypes and clonality

A total of 278 invasive GBS isolates were analyzed: 133 (47.8%) from EOD patients and 145 (52.2%) from LOD patients. Among the 278 isolates, 61.5% (171) were from blood only, 9.0% (25) were from CSF only, and 29.5% (82) were isolated from both blood and CSF. The predominant serotypes were III (n=163; 58.6%) and Ib (n=75; 27.0%); four serotypes (Ia, Ib, III, V) accounted for 97.8% (272) of all isolates; and 2 (0.72%) were non-typeable. The percentage of serotypes differed between cases of EOD and LOD; 49.6% (66/133) of the EOD isolates were serotype III, while 66.9% (97/145) of the LOD case isolates were serotype III (Fig.1A).

We revealed 30 sequence types that were grouped into five CCs (CC1, CC10, CC17, CC19, and CC23) (Fig.2). The predominant CC was CC17 (113/278; 40.6%), and the majority of GBS cases (264/278; 95.0%) were attributable to CC10, CC17, CC19, and CC23 isolates (Fig.1B). Most serotype III (112/163; 68.7%) and Ib isolates (74/75; 98.7%) belonged to CC17 and CC10, respectively (Fig.1C). Furthermore, we identified six isolates not belonging to the five typical CCs: one was ST103, two were ST651, and three were ST862. The 6 isolates had similar phylogenetic origins, and based on their allele identities and SNP alignments (Fig.2), they were clustered with the ST485 strain ROI138, which was isolated from blood samples of a female patient in Ireland. Considering that ST485 had been identified as CC327 in pubMLST (https://pubmlst.org/organisms/streptococcus-agalactiae) and that ST862, ST103, and ST651 had triple-allele, four-allele, and 5-allele variants, respectively, with ST327 as the founder sequence type of CC327, we named a new CC651 clone cluster for the three sequence types (ST103, ST651, and ST862), in which ST103 and ST862 were single-allele variants and double-allele variants, respectively, with ST651 (Fig.2 and Supplementary Fig.1).

3.2 Surface protein genes and virulence factors

A total of 98.9% (275/278) of the isolates harbored at least one alpha-like protein-encoding gene. The *rib* gene was the most common gene detected (n=155; 55.8%). The *rib* gene was detected in only serotypes III (n=152; 98.1%), Ib (n=1; 0.6%), and V (n=2; 1.3%). Furthermore, the prevalence of other alp-like proteins was 28.8%, 12.2%, and 2.2% for *alpha* C (n=80), *alp1* (n=34) and *alp2/3* (n=6), respectively. Most *alpha C*-positive isolates were serotype Ib (74/80; 92.5%), and the majority (28/34; 82.4%) of the *alp1* isolates were serotypes Ia and III (Table 1, Fig.3, Fig.4 and Supplementary Fig.2).

All invasive GBS isolates contained at least one of the three pilus backbone determinants (Table 1 and Fig. 3). The most prevalent pilus types were *PI-2b* (n=111; 39.9%) and *PI-1+PI-2a* (n=111; 39.9%), followed by *PI-2a* (n=49; 17.6%). The type of pilus island was strongly correlated with the ST and CC. Most CC17 (105/113, 92.9%) and all CC651 (6/6, 100%) were *PI-2b*. Ninety-five percent of CC19 (52/55) and 68% of CC12 (51/75) were *PI-1+PI-2a*. All the ST23 (19/19, 100%) was *PI-2a* (Table 2).

All 111 serotype III/CC17 GBS isolates were positive for *hvgA*. Most of the isolates (209/278, 75.2%) were positive for serine-rich repeat glycoprotein determinants (*srr1* or *srr2*). Almost all *srr2* genes were found in CC17 (112/113, 99.1%) isolates. In contrast, srr1 genes were found in only other CCs (Table 2 and Fig. 3). cfb, c1IE, lmb and pavA were the most common virulence genes identified in 99.6%

(277/278), 100% (278/278), 95.3% (265/278) and 100% (278/278) of the isolates, respectively (Fig. 3, Supplementary Table 1, Supplementary Table 2 and Supplementary Fig. 3). The h1lb gene was detected only in 150 (53.9%) isolates. In comparison, *fbsA* was detected only in 2 (0.7%) isolates.

3.3 Antimicrobial resistance determinants

Twenty-two antimicrobial resistance genes were identified in at least one of the 278 invasive GBS genomes (Fig. 3 and Supplementary Table 3). Among the 278 isolates, 1.1% (3), 22.3% (62), and 76.6% (213) contained 1, 2 and 3 resistance genes, respectively. Notably, almost all the isolates carried the macrolide nonsusceptibility gene *mreA* (277/278), while 72.7% (202), 28.1% (78), and 28.1% (78) had *ermB*, *mefA,* and *msrD*, respectively. The tetracycline nonsusceptibility genes *tetO* and *tetM* were prevalent in 55.4% (154/278) and 42.8% (119/278) of the CC17 clones, respectively. The *lnuB* gene associated with clindamycin antibiotic resistance was found in 17.7% (20/278) of CC17 and 27.8% (15/278) of CC19 strains, while 4 resistance genes *(ant(6)-Ia, aph(3')-III, aph(2'')-Ia* and *aac(6')-aph(2''))* associated with aminoglycoside nonsusceptibility were found in more than 50% of the isolates and mostly in CC17 or CC10 clones.

SNPs in *gyrA* and *parC* were the most common fluoroquinolone nonsusceptibility mutations and were present in 79 and 83 isolates, respectively; 74.1% of *gyrA* isolates and 75.9% of *parC* isolates were CC19 isolates and CC10 isolates, respectively (Fig. 3, Supplementary Table 3 and Supplementary Fig. 4). Notably, there were no determinants associated with nonsusceptibility to penicillin. In the drug resistance gene screening analysis, one isolate was found to carry the oxazolidinone resistance gene *optrA*; however, the isolate did not show a resistance phenotype to oxazolidinone drugs. Further sequence alignment analysis of the *optrA* gene in the isolate revealed that it lacks the ATG start codon and its subsequent 24 nucleotides. Consequently, the gene could not be properly expressed, nor did it exhibit the corresponding resistance phenotype.

4. Discussions

This nationwide multicenter, population-based surveillance study reported the largest WGS analysis of GBS isolates collected from Chinese infants with invasive disease. We revealed the distributions of candidate GBS vaccine targets, including capsule polysaccharides and surface proteins. We found that current GBS vaccine candidates cover more than 98% of invasive disease cases in this study. Penicillin remains appropriate for first-line prophylaxis and treatment, but multiple resistance leads to fewer options for GBS patients with penicillin allergy. Continuous surveillance of resistance features is therefore warranted.

The presence of hypervirulent genomic regions was predictably associated with dominant GBS serotypes. We reported a predominance of CC17, CC19, and CC23 in the Chinese population, which is largely consistent with global findings, and the distribution of CCs was comparable to that in other regions[19, 26, 27]. ST103, ST651, and ST862 were first reported in pregnant women in 2018, with no evidence of infection [28]. In contrast, we showed that these three sequence types caused GBS EOD and LOD among infants. The detection of ST103, which is common in GBS animal sources and rarely found in human isolates [29, 30], suggests that more attention should be given to preventing cross-host infection in humans with GBS animal sources.

Several candidate capsular polysaccharide conjugate GBS vaccines have undergone clinical trials[16, 31, 32]. Pentavalent GBS vaccines, including serotypes Ia, Ib, II, III, and V, or hexavalent (serotypes Ia, Ib, II, III, IV, and V) GBS conjugate vaccines [16, 33], accounted for the majority (98%) of invasive disease cases in our study.

In addition to CPS-conjugate vaccines, the use of GBS surface proteins as vaccine candidates could broaden protection against invasive GBS infections [10]. Genes encoding potential GBS protein vaccine epitopes, such as pilus antigens, *srr1,* and *ssr2,* as well as *alp proteins,* were identified in a substantial number of our study isolates. A prototype subunit GBS-NN vaccine containing the N-terminal domains of AlphaC and the *rib* displayed good safety and immunogenicity [34]. Furthermore, a second-generation vaccine, a two-component vaccine containing the four Alp N-terminal domains AlphaC-N, Rib-N, Alp1-N, and Alp2/3-N, has been developed [35, 36] and has the potential to prevent up to 99% of invasive disease in infants in the Chinese population. In addition to the Alp protein-based vaccine, pilus proteins have also been proposed as potential vaccine candidates [37] that could target all GBS proteins in this study.

The pilus protein *PI-2b* is a CC17- and CC651-specific virulence factor in this study that may enhance GBS adhesion, leading to the development of meningitis[38]. Our results also confirmed that *hvgA* was found only in CC17 isolates, while 99.1% of the *srr2* genes were found in CC17 isolates. The *cfb-cylE-lmb-pavA* pattern was the most dominant virulence gene profile mainly found in CC17 in this study.

Penicillin G is recommended as the first-line agent, whereas cefazolin, clindamycin, vancomycin, and fluoroquinolones are recommended as alternative medications for individuals with known allergies to penicillin in China. Currently, susceptibility to β-lactams and vancomycin is very rare in the Chinese population. However, focused monitoring of diverse resistance features is important given the rapidly growing proportion of invasive GBS strains that are resistant to macrolides and lincosamides, which are conferred by diverse genetic determinants[39, 40]. Twenty-two antimicrobial resistance genes were identified in this study, and 76.6% (213) of the 278 isolates contained three resistance genes. Multiple resistance to erythromycin, levofloxacin, and tetracycline leads to fewer antibacterial options available for GBS patients with penicillin allergy. Furthermore, given the continued accumulation of SNP mutations and environmental selection pressure, GBS strains with phenotypic resistance to other potential antimicrobial agents might emerge in the future. Therefore, continuous monitoring of resistance features among GBS isolates is essential to control the rapid increase in GBS antimicrobial resistance[3].

The limitations of this study include that the GBS isolates were obtained from large urban tertiary hospitals because of the lack of adequate laboratory facilities for GBS culture and identification of invasive GBS infections at secondary hospitals and primary care institutions. However, compared to previous single-center or regional studies in China, we included different geographical regions that were also distinct in terms of GDP to obtain a more representative sample. Second, the GBS isolates and data collected from 2018 to 2021 were only available from some of the selected hospitals..

5. Conclusions

In conclusion, this is the largest WGS analysis of invasive GBS isolates among Chinese infants. Current vaccine targets would have good coverage in China. The genotype, surface proteins, virulence factors, and resistance genes identified in our multicenter population-based study provide key insights into GBS vaccine development and treatment. These findings establish important baseline data required for further surveillance and evaluating the impact of future vaccine candidates on the molecular epidemiology of GBS.

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Credit author statement

WJ, HZ, JL, and YF conceptualized the study. WJ, HZ, and JL designed the study. WJ, JL, JD, and CDB analyzed the data. WJ drafted the manuscript. HZ, JL, and CDB revised the manuscript. ZL, WZ, JL, JD, HZ, YZ and ZJ contributed to the laboratory work and acquisition of the data. SAM, GK, and ZD provided critical comments and suggestions for the manuscript. All authors reviewed and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no conflicts of interest.

All funders had no role in the study design, data collection and analysis, manuscript writing, or the decision to submit the manuscript for publication. All authors attest that they meet the ICMJE criteria for authorship.

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Table 1. Surface protein determinants in relation to serotypes among invasive GBS disease isolates.

	Surface protein determinants										
Serotype	srr1	srr2	hvgA	PI-2a	PI-2b	PI-1+PI-2a	PI-1+PI-2b	alp1	alp2/3	Alpha C	rib
Ia (n=21)	19 (90.5%)	0 (0%)	0 (0%)	18(85.7%)	0 (0%)	3(14.2%)	0 (0%)	18 (85.7%)	2 (9.5%)	0 (0%)	0 (0%)
Ib (n=75)	43 (57.3%)	0 (0%)	0 (0%)	24(32%)	0 (0%)	51(68%)	0 (0%)	0 (0%)	0 (0%)	74 (98.7%)	1 (1.3%)
II (n=1)	1 (100%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (100%)	0 (0%)	0 (0%)	0 (0%)	1 (100%)	0 (0%)
Ⅲ(n=163)	20 (12.3%)	112 (68.7%)	111 (68.1%)	1(0.6%)	110 (67.4%)	45 (27.6%)	7(4.2%)	10 (6.1%)	0 (0%)	0 (0%)	152 (93.3%
Ⅳ(n=1)	0 (0%)	1 (100%)	0 (0%)	0 (0%)	1 (100%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (100%)	0 (0%)
Ⅴ(n=13)	11 (84.6%)	0 (0%)	0 (0%)	3 (23%)	0 (0%)	10 (76.9%)	0 (0%)	6 (46.2%)	4 (30.8%)	1 (7.7%)	2 (15.4%)
Ⅵ(n=2)	2 (100%)	0 (0%)	0 (0%)	1 (50%)	0 (0%)	1 (50%)	0 (0%)	0 (0%)	0 (0%)	2 (100%)	0 (0%)
NT (n=2)	0 (0%)	0 (0%)	0 (0%)	2 (100%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (50%)	0 (0%)
Total (n=278)	96 (34.5%)	113 (40.6%)	111 (39.9%)	49 (17.6%)	111 (39.9%)	111 (39.9%)	7 (2.5%)	34 (12.2%)	6 (2.2%)	80 (28.8%)	155 (55.8%

Table 2. Surface protein determinants in relation to clonal complex among invasive GBS disease isolates.

Clonal Complex	Surface protein determinants										
	srr1	srr2	hvgA	PI-2a	PI-2b	PI-1+PI-2a	PI-1+PI-2b	alp1	alp2/3	Alpha C	rib
CC1 (n=8)	8 (100%)	0 (0%)	0 (0%)	1 (12.50%)	0 (0%)	7 (87.5%)	0 (0%)	1 (12.5%)	4 (50.0%)	3 (37.5%)	0 (0%)
CC10 (n=75)	43 (57.3%)	0 (0%)	0 (0%)	24(32%)	0 (0%)	51 (68%)	0 (0%)	0 (0%)	0 (0%)	75 (100%)	0 (0%)
CC17 (n=113)	0 (0%)	112 (99.1%)	111 (98.2%)	1(0.8%)	105 (92.9%)	0 (0%)	7 (6.1%)	0 (0%)	0 (0%)	0 (0%)	112 (99.1%
CC19 (n=54)	20 (37.0%)	0 (0%)	0 (0%)	3 (5.5%)	0 (0%)	51(94.4%)	0 (0%)	10 (18.5%)	0 (0%)	0 (0%)	43 (79.6%)
CC23 (n=22)	19 (86.4%)	1 (4.6%)	0 (0%)	20 (90.9%)	0 (0%)	2(9%)	0 (0%)	18 (81.8%)	2 (9.1%)	2 (9.1%)	0 (0%)
CC651 (n=6)	6 (100%)	0 (0%)	0 (0%)	0 (0%)	6 (100%)	0 (0%)	0 (0%)	5 (83.3%)	0 (0%)	0 (0%)	0 (0%)
Total (n=278)	96 (34.5%)	113 (40.6%)	111 (39.9%)	49 (17.6%)	111 (39.9%)	111 (39.9%)	7 (2.5%)	34 (12.2%)	6 (2.2%)	80 (28.8%)	155 (55.8%

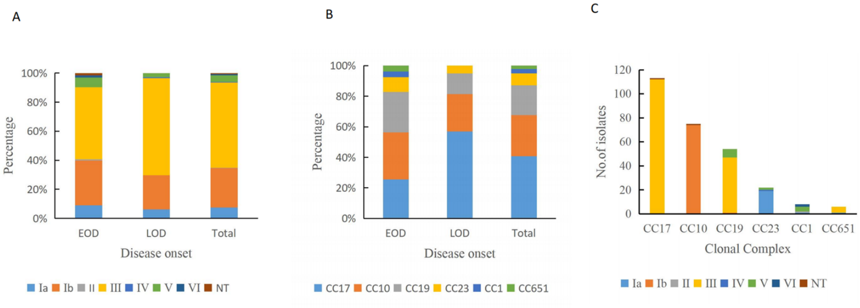
Figure legends

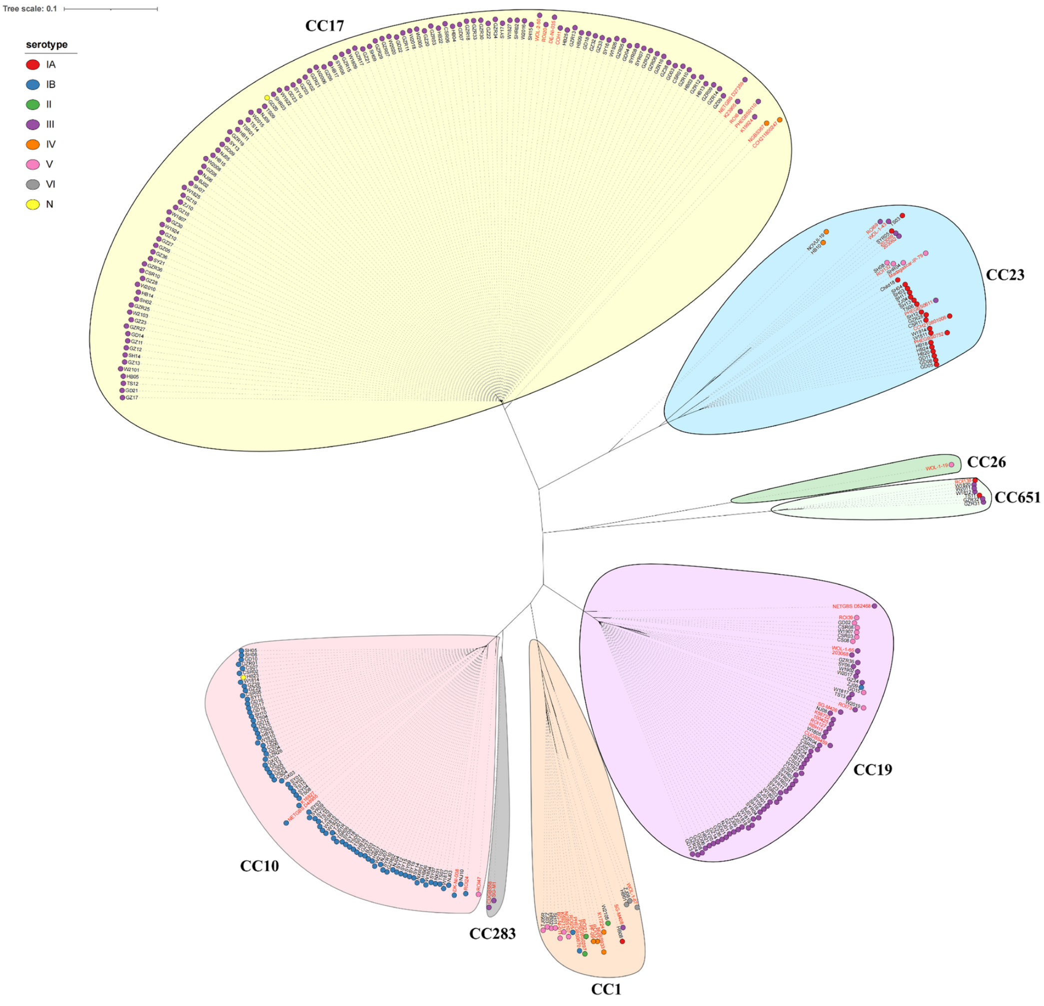
Fig. 1. A, Serotype distributions of invasive group B *Streptococcus strains stratified* by disease onset. B, Clonal complex (CC) distributions by disease onset. C, Serotype distribution of each CC

Fig. 2. The phylogenetic tree was constructed by SNP alignment. The total number of group B *Streptococcus* isolates in this figure was 333, including 278 isolates from this study, 49 worldwide representative invasive disease isolates, and 6 isolates representative of 5 main cluster clones distributed worldwide and a CC26 representative strain unique to African strains downloaded from PubMLST (https://pubmlst.org/organisms/streptococcus-agalactiae) and NCBI. The isolates marked in red and black were downloaded from PulMLST and NCBI, respectively, and used in this study.

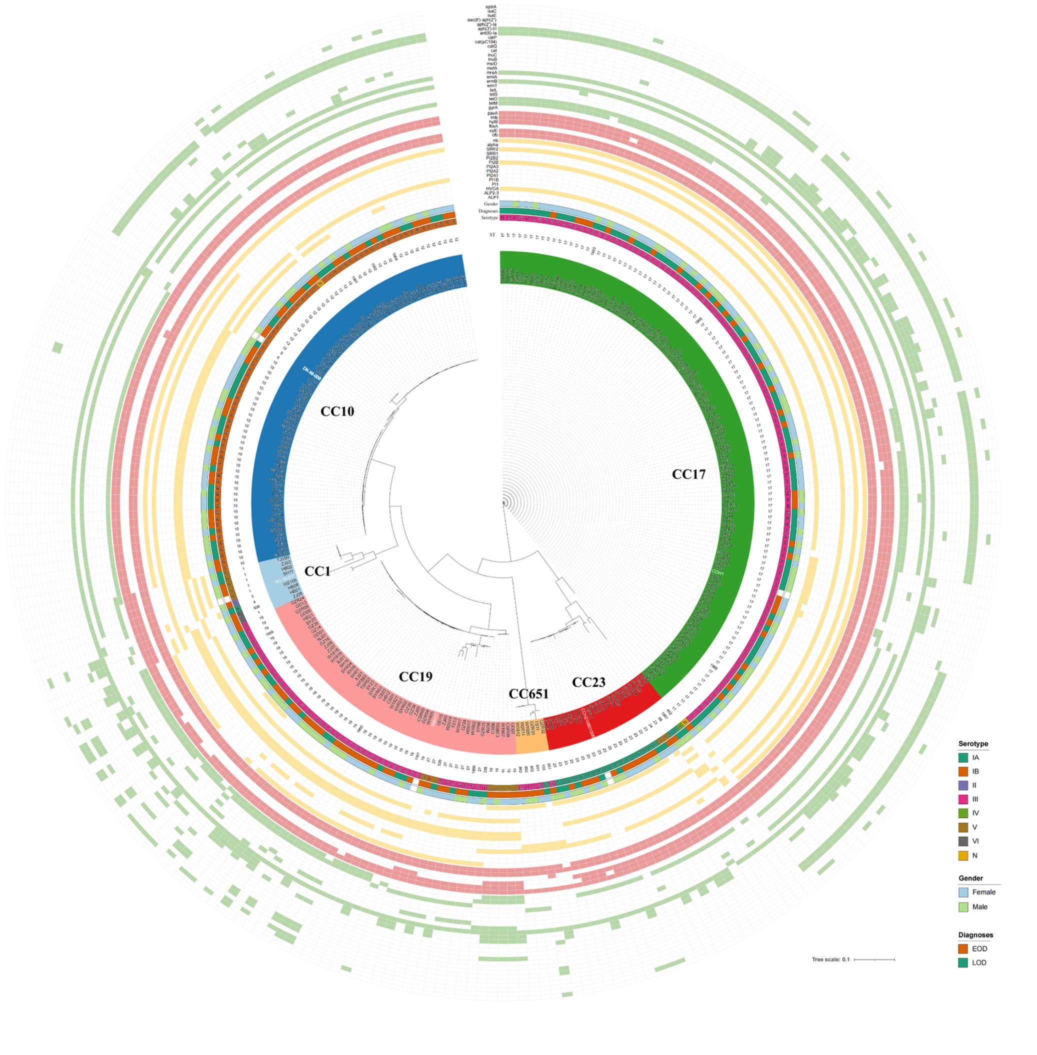
Fig. 3. Whole-genome SNP-based maximum-likelihood phylogenetic tree and heatmap of 278 group B *Streptococcus* strains showing virulence and resistance genes. The spaces are filled in yellow, red, or green if the isolate harbors the surface protein genes, virulence factors, or resistance genes, respectively.

Fig. 4. The phylogenetic tree of 278 group B *Streptococcus* isolates and 5 reference strains constructed based on the tandem combinations of 11 surface proteins (*alp1, alp2/3, alpC, C5a, gbs2106, PI-1, PI-2a, PI-2b, rib, sip* and *srr2)*. The rings from inner to outer represent the serotypes, clone complexes, and sequence types (STs). The 5 reference strains are indicated in yellow.

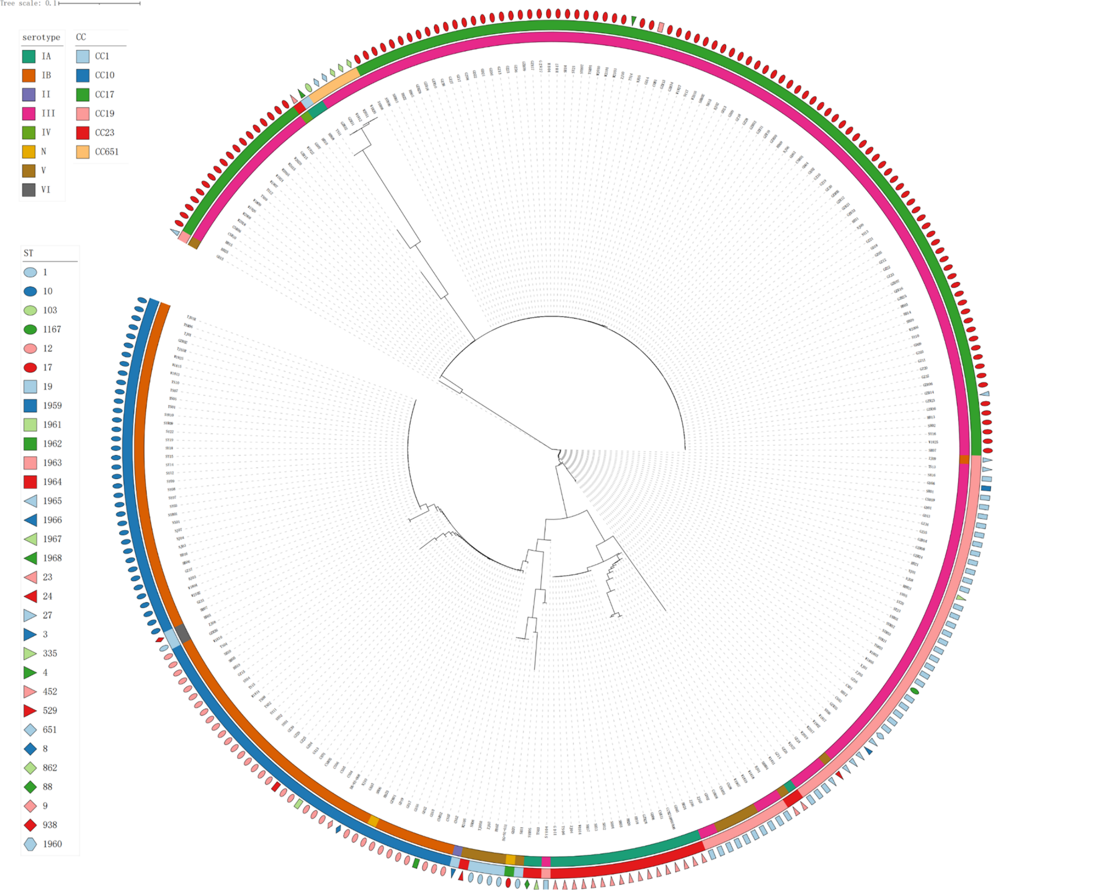

A
100%
80%
60%
40%
20%
0%
Percentage
EOD
LOD
Total
Disease onset
Ia Ib II III IV V VI NT
B
100%
80%
60%
40%
20%
0%
Percentage
EOD
LOD
Total
Disease onset
CC17 CC10 CC19 CC23 CC1 CC651
C
120
100
80
60
40
20
0
No.of isolates
CC17 CC10 CC19 CC23 CC1 CC651
Clonal Complex
Ia Ib II III IV V VI NT



Tree scale: 0.1
serotype
IA
IB
II
III
IV
V
VI
N
CC17
CC23
CC26
CC651
CC19
CC10
CC283
CC1



optrA
lsaC
lsaE
aac(6')-aph(2'')
aph(2'')-Ia
aph(3')-III
ant(6)-Ia
catP
cat(pC194)
catQ
cat
lnuC
lnuB
msrD
mefA
mreA
ermA
ermB
ermT
tetL
tetS
tetO
tetM
gyrA
pavA
lmb
hylB
fbsA
cylE
cfb
rib
alpha
SRR2
SRR1
PI2B2
PI2B
PI2A3
PI2A2
PI2A1
PI1B
PI1
HVGA
ALP2-3
ALP1
Gender
Diagnoses
Serotype
ST
CC10
CC1
CC19
CC651
CC23
CC17
Serotype
IA
IB
II
III
IV
V
VI
N
Gender
Female
Male
Diagnoses
EOD
LOD
Tree scale: 0.1


Tree scale: 0.1
serotype
IA
IB
II
III
IV
N
V
VI
CC
CC1
CC10
CC17
CC19
CC23
CC651
ST
1
10
103
1167
12
17
19
1959
1961
1962
1963
1964
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1967
1968
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1960